

countering the distortion caused by the viewer's lenses, and ultimately what you get is a beautiful stereoscopic, three-dimensional (3D) image with a wide field-of-view (FOV)!

Makes a lot of sense, does it not? When your mobile phone has everything you need including the processor and the sensors, why not use it with an appropriate accessory for VR, too? It is this logic that has brought VR to the masses today.

Indeed, Google Cardboard is fast growing into a popular VR platform, including options for developing apps as well as creating 360-degree video content. Third-party products are also being made Cardboard-compatible, so that users can benefit from the exciting apps available on Google Play. In March, Google improved accessibility by adding iOS support to Cardboard. They also announced that they will be making a VR viewer that can work independently without a smartphone or a computer. This is expected to be a sturdy, plastic-encased viewer.

Asked about the effectiveness of low-cost Cardboard based viewers, Lalpura says, "In some aspects, these are as good as the pure-bred VR solutions. But, obviously, if you are going to compare, say, Google Cardboard with Samsung Gear VR, then you can see the difference. The quality of the headset body and lenses (for example, FOV) also matters. But for basic users, Google Cardboard can definitely do the job."

If you want something sturdier than the current Cardboard, you can go for some of the Cardboard-inspired and Cardboard-compatible products like Goggle Tech's Go4D C1-Glass and Homido Mini.

Mattel's View-Master is also Cardboard-compatible. There are other options like Powis ViewR 2.0 that are slightly costlier (around US\$ 30) but include substantial features to justify the cost.

ViewR 2.0, for example, has lenses that adjust independently to

What is new

- Low-cost frames and apps that turn your mobile phone into a VR headset
- Better refresh rates to reduce nausea
- Improved displays including OLED
- Position-tracking systems that turn your room into a 3D space, so you can move around your virtual world
- Handheld controllers to manipulate objects in the virtual space
- Eye-tracking capabilities for more natural views
- Virtual environment made of high-definition holograms
- Open source designs and development kits

correct several vision problems like far-sightedness and astigmatism. It works with 10.2cm to 16.5cm (4-inch to 6.5-inch) Android, iPhone and Windows phones.

Homido VR Headset, which works with most smartphones, is another interesting headset with a 100-degree FOV. It too has adjustable lenses and different modes for different types of sight.

FreelyVR steps up the FOV to 120 degrees. It also has a Bluetooth remote and head-tracking technology built into it. However, it does not have adjustable lenses, so it is not for those with vision problems.

Samsung Gear VR is acclaimed as another fabulous option in the sub-US\$ 100 range, but it, unfortunately, works only with Samsung phones.

If you loosen the purse strings a bit more, you can go for options like Zeiss VR One, a VR headset made by a company known for its expertise in optics. Priced around US\$ 130, it works with iPhone 6, Samsung Galaxy S4, S5, S6, Nexus 5 and LG-G3 smartphones, and is positioned as a premium product in terms of style and comfort.

Full-fledged but still within reach

The real VR viewers are also thriving, with innumerable tech advancements and new products being launched. Affordability is a trend in this segment, too. Gone are the days when you had to shell out US\$ 2000 or more to get a proper VR experience. Priced between US\$ 500 and

US\$ 1000, the latest generation of gadgets like HTC Vive and Oculus Rift offer amazing value for money with mind-blowing features.

Three of the most talked about VR launches of this year are Valve's HTC Vive, Sony's PlayStation VR and Facebook's Oculus Rift. Overall, this generation features a 100+ FOV, 360-degree motion tracking, high resolution and an average refresh rate of around 90Hz, which apparently helps reduce nausea. Most of these products also include handheld controllers with motion-tracking capabilities and haptic feedback, so that users can lift, move and manipulate objects in their virtual environment.

Products like Vive and Rift also attempt to turn your room into a 3D space, so that you can move around your virtual world. While Valve's position tracking system is called SteamVR and uses two base stations, Oculus' tech is called Constellation and uses a single table-top sensor. Basically these base stations keep scanning the room with light—laser and infrared in the case of SteamVR; infrared and light-emitting diode (LED) in the case of Constellation.

A series of infrared diodes are also studded on the headset and handheld controller. Time and other position-tracking parameters are used to calculate the position of these objects relative to the known position of the base station. This, in turn, enables the user to walk around and move objects in their virtual world.

Sony's PlayStation VR, which is expected to be launched later this